Giacomo Pasin

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Education

ETH Zürich Sep 2026 – Present
MSc in Computer Science *Zürich, Switzerland*

- Major: Machine Intelligence; Minor: Data Management Systems

Eindhoven University of Technology Sep 2023 – Jul 2026
BSc in Computer Science and Engineering *Eindhoven, Netherlands*

- Graduated *Cum Laude*
- Thesis: *Explainable RF Fingerprinting*. Reproduced and evaluated two spectrogram-based ML pipelines and conducted a partial ablation study to investigate misconfiguration.

Experience

AI Engineering and Software Development Intern Jun 2026 – Aug 2026
Mea World *Pistoia, Italy*

- Improved source selection in the RAG pipeline behind Kiuwo, so answers draw on more relevant documents
- Developed an internal AI agent that autonomously resolves the tickets assigned to it and reviews its own output against the original spec

CTO Aug 2021 – Present
My Twin Communication *Treviso, Italy*

- Consolidated infrastructure across two merging communication agencies and built internal tools supporting workflows across departments.

Foundations of Data Analytics Development Team Nov 2024 – May 2026
TU Eindhoven *Eindhoven, Netherlands*

- Built Jupyter notebook test frames and exercises for the course "Foundations of Data Analytics":
- Wrote test cases for the 1000+ student submissions

Selected Projects

CheckInOS | *TypeScript, Next.js, tRPC* 02/2025

- Enabled guest check-in on two mobile platforms (iOS and Android) by connecting native QR scanning to web-managed attendee rosters.
- Delivered real-time event operations with live registration and arrival analytics, Stripe ticketing, and on-site badge printing in one workflow.
- Built AI-powered feedback analysis that generates event reports from form responses, adapting report structure and insights to each form's questions and answer types to surface strengths and weaknesses.

ML for IR Spectroscopy | *Python, PyTorch, OpenCV* 06/2025

- Built a training corpus of 60,000+ IR spectra in under two days by developing two Selenium scrapers for NIST and SDBS and normalizing GIF and JCAMP-DX data.
- Extended functional-group prediction from pure compounds to binary mixtures by reproducing a published CNN in PyTorch and developing a fine-tuning pipeline using synthetic mixture spectra.
- Enabled spectrum uploads to return functional-group probabilities through a web interface by deploying the trained model behind a Flask API with Swagger documentation.

Research Terminal | *TypeScript, Next.js, Claude* 09/2026

- Enabled literature discovery through three entry points (title, DOI and natural-language ideas) by combining keyword and semantic search over OpenAlex.
- Made literature reviews resumable, with saved multi-paper citation graphs and per-paper AI summaries, by integrating reference/citation expansion, Claude summarization and account-based autosave.

Skills

ML & Data: Python, PyTorch, NumPy, scikit-learn, Pandas, OpenCV, Matplotlib, Jupyter Notebooks
Technologies: TypeScript, Git, Docker, Linux, Redis, MongoDB
Languages: English (Native), Italian (Native), Spanish (B2)